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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Subject: Operating System

Unit : 1

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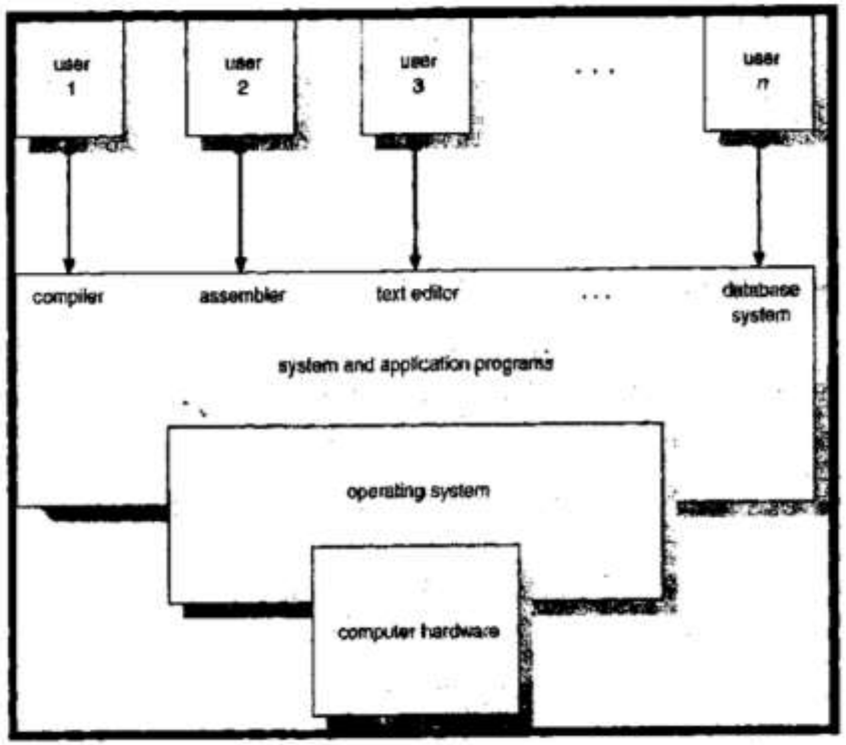
INTRODUCTION TO OPERATING SYSTEM

- An **operating system** is a program that manages the computer hardware. It also provides the basis for application programs and acts as an intermediary between the user of a computer and the computer hardware. It controls and coordinates the use of the hardware among the various application programs for the various users.

Operating Systems can be explored from two viewpoints.

1) **User View:** The user view of the computer varies by the interface being used. Single-user systems are designed to maximise the work the user is performing; in this case operating system is designed mostly for ease of use, performance and less resource utilisation. Multiuser systems share resources and exchange information; in this case operating system is designed to maximise resource utilisation.
2) **System View:** From the computer's point of view, the operating system is a Program that is most intimate with the hardware and can be viewed as a resource allocator. Another view of the operating system is, as a control program that controls various I/O devices and user programs.



ABSTRACT VIEW OF SYSTEM COMPONENTS

BATCH OS

- The computer runs one and only one application. Its major task was to transfer control automatically from one job to the next. The operating system is always resident in the memory.
- In Batch systems, the operator will sort programs into batches with similar requirements and run each batch to speed up the processing. For the batch systems, most of the time, the CPU is idle because of the low speeds of mechanical devices.
- The introduction of disk technology allowed the operating system to keep all jobs in disk rather than in a card reader. With direct access to jobs, the operating system performs job scheduling, to use resources and perform tasks efficiently.

Multi-programmed System:

- A single-user system cannot keep the CPU or the I/O devices busy at all times. To increase CPU utilisation, the multi-programmed systems use the concept of job scheduling
- The CPU always has one to execute.
- The operating system keeps several jobs in memory simultaneously.
- This set of jobs is a subset of the job pool. The operating system picks and begins to execute one of the jobs in memory.

TIME SHARING SYSTEM

- Time-sharing (or multitasking) is a logical extension of multiprogramming. The CPU switches among multiple jobs for execution, but the switching occurs so frequently. So, the user can interact with the system while the program is running.
- A time-sharing operating system allows multiple users to share the system simultaneously, as the system switches from one user to the next; each user is given an impression that the entire computer system is dedicated to a single user, even though it is being shared among many users. Time-sharing systems are more complex than multiprogramming systems.

Multiprocessor systems

- Most of the systems are single-processor systems; that is, they have only one main CPU. However, multiprocessor systems, also known as parallel systems or tightly coupled systems, have more than one processor and share the computer bus, the clock, and the peripheral devices and memory often.
- Multiprocessor systems have three advantages. They are :

1. **Increased throughput:** By increasing the number of processors, more work gets done in less time.
2. **Economy of scale*** Multiprocessor systems save money than multiple single processor systems because they can share peripherals, storage, data and power supplies.
3. **Increased reliability:** The distribution of functions among various processors will restrict the system from a halt state even though one processor has failed.

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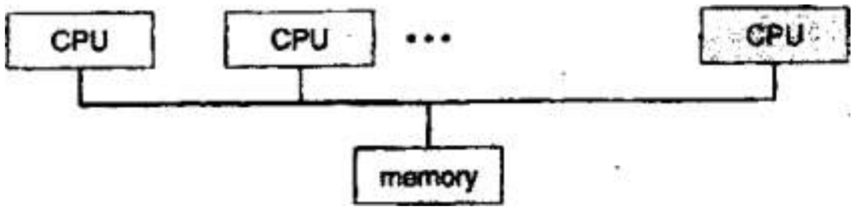
SMP means that all processors are peers; no master-slave relationship exists between processors. The fig illustrates SMP architecture.

- SMP have multiple processors, so one may be sitting idle, and another is overloaded, which results in inefficiencies. These can be avoided by sharing data structures among the processors.
- A multi-processor system of this form will allow processes and resources such as memory to be shared dynamically among the various processors and can lower the variance among the processors.
- In Asymmetric multiprocessing, the master-slave relationship scheme is defined. The master processor controls and allocates work to the slave processors. Each processor is assigned a specific task. The slave processors respond to a master processor's instructions or they have predefined tasks.



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Distributed System

- Distributed systems are able to share computational tasks, and provide a rich set of features to users by distributing their functionality over networking for communication.
- A network is the communication path between two or more systems. Networks vary by the protocols used, the distance between the nodes, and the transport media. Most operating systems support TCP/IP, including the Windows and UNIX OS.

Client- Server systems

- As PCs become faster, more powerful and cheaper, designers have shifted away from the centralised system architecture to client-server systems.
- The server system satisfies the requests generated by the client systems. The general structure of the client-server system is depicted in the figure.

Server

Server systems can be broadly categorised as:

• **Compute-server systems:** provides an interface to which clients can send requests to perform an action, and then the server responds to it by executing the action and sending back the results to the client.

REAL-TIME SYSTEMS

- Real-time constraints have been placed on the operation of a processor or the flow of data. So, it is often used as a control device in dedicated applications.
- Systems that control scientific experiments, medical imaging systems, industrial control systems, and certain display systems are real-time systems.
- Real-time systems have well-defined, fixed time constraints. The processing must be done within the defined constraints, or the system will fail.
- A real-time system functions correctly only if it returns correct results within its time constraints.
- Real-time systems are two types. They are -
- Stem is another form of special-purpose OS. A real-time system is used when rigid time.

  a. **Hard real-time system:** This guarantees the critical tasks to be completed on time.

  b. **Soft real-time systems:** A less restrictive type of real-time System, where a critical task gets priority over other tasks and retains priority until it finishes the task.

COMPUTER SYSTEM STRUCTURES

- A general-purpose computer system consists of a CPU and a number of device controllers that are connected through a common bus that provides access to shared memory. Each device controller controls a specific type of device like Disk drives, Audio devices, and video displays.
- A memory controller is provided to synchronise access to memory that is shared by the CPU and device controllers.
- When the computer is powered up or rebooted, it needs an initial program to start running the system, i.e. Bootstrap program. It is stored in Read-only memory (ROM). It initialises all CPU registers to device controllers.
- The bootstrap program must load the operating system and start executing the first process, such as "init," and wait for some event signalled by interrupts from either the hardware or the software.
- Software may interrupt the execution of a special operation called a system call (also called a monitor call).
- A trap (or an exception) is a software-generated interrupt caused either by an error like division by zero or invalid memory access.

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- The interrupt routine is called through a table of pointers, which holds the address of the interrupt service.

Routines of the various devices.

- If the interrupt routine needs to modify the processor state, it must save the current state and restore that state before returning to interrupted computation.

I/O structure

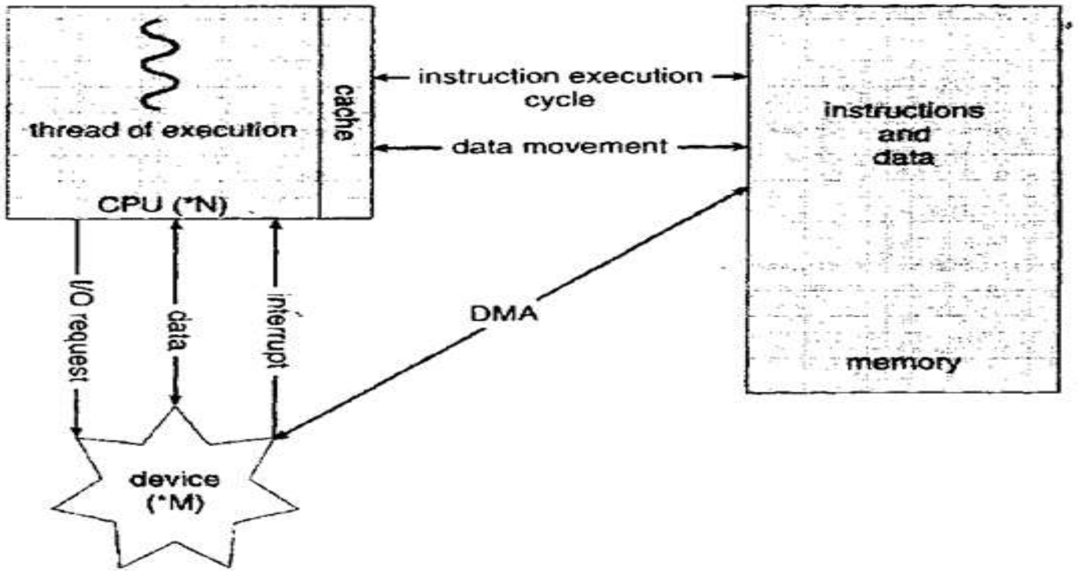
- Storage is one of the many types of I/O devices within a computer. A large portion of operating system code is dedicated to managing I/O because of its importance to the reliability and performance of a system.
- To start an I/O operation, the CPU loads the appropriate registers within the device controllers. When the I/O is started, two courses of action are possible.

I/O Interrupts

- Once the I/O is started, two courses of action are possible. In the simplest case, the I/O is started then, at completion of the I/O, control is returned to the user process. This case is called synchronous I/O. Waiting for I/O completion is done in two ways.
- By wait instruction that idles the CPU until the next interrupt.
- By the wait loop for machines that don’t have wait instructions. This loop continues until an interruption has occurred.

DMA STRUCTURE

- This form of interrupt-driven I/O is fine for moving small amounts of data, but it is overhead for bulk data movement such as disk I/O. To solve this problem, direct memory access (DMA) is used. After setting up buffers, pointers and counters for the I/O devices, the device controller transfers an entire block of data directly to or from its own buffer storage to memory, with no intervention by the CPU.
- Only one interrupt is generated per block rather than per byte. While the controller is performing these tasks, the CPU is available to perform other tasks. The DMA controller interrupts the CPU when the transfer has been completed.



STORAGE STRUCTURE

- The permanent storage of data and programs in main memory is not possible because of limited main memory, and main memory is a volatile storage device. It loses its contents when power is switched off. RAM is the storage area that the processor can access directly. Interaction with RAM is achieved by loading and storing instructions. The load instruction moves data from the main memory to the internal register, whereas the store moves the content of a register to the main memory.
- A typical instruction execution cycle will first fetch instruction from memory and store that instruction in the instruction register. The instruction is then decoded, which causes the fetch operands to execute the instruction and store the result back to memory.
- So, most of the computers had secondary storage as an extension to main memory. The most commonly used secondary storage device is a magnetic disk, which provides storage of both programs and data.
- Many programs use disk as both a source and a destination of the information for their processing. There are various other storage systems like cache memory, CD-ROM, magnetic tapes and so on. The main difference among them lies in the speed, cost, size and volatility.

STORAGE STRUCTURE

Magnetic Disks-

- Magnetic disks provide the bulk of secondary storage for modem computers. Each disk platter has a flat circular shape like CD. Common platter diameter ranges from 1.8 to 5.25 inches. The two surfaces of the platter are covered with a magnetic material. A read-write head “flies” just above each surface of every platter.
- The heads are attached to a disk arm, which moves all the heads as a unit. The surface of a platter is logically divided into circular tracks, which are subdivided into sectors. The set of tracks that are at one arm position forms a cylinder.
- The storage capacity of common disk drives is measured in gigabytes. When the disk is in use, the disk motor rotates at 60 to 200 times per second.

Magnetic Tapes

- Magnetic tape was used as an early secondary storage medium. Magnetic Tapes can hold large quantities of data, and it is permanent.
- The access time of the magnetic tape is slower compared to the main memory and magnetic disk.
- The tape is kept in a spool and is wound or rewound past a read-write head. Moving to the correct position on a tape takes minutes, but write data at speeds comparable to disk drives once placed in the correct position.
- The capacity of tape drives varies based on the width of the drives. Magnetic Tapes are mainly used for backup purposes and transferring information from one system to another.

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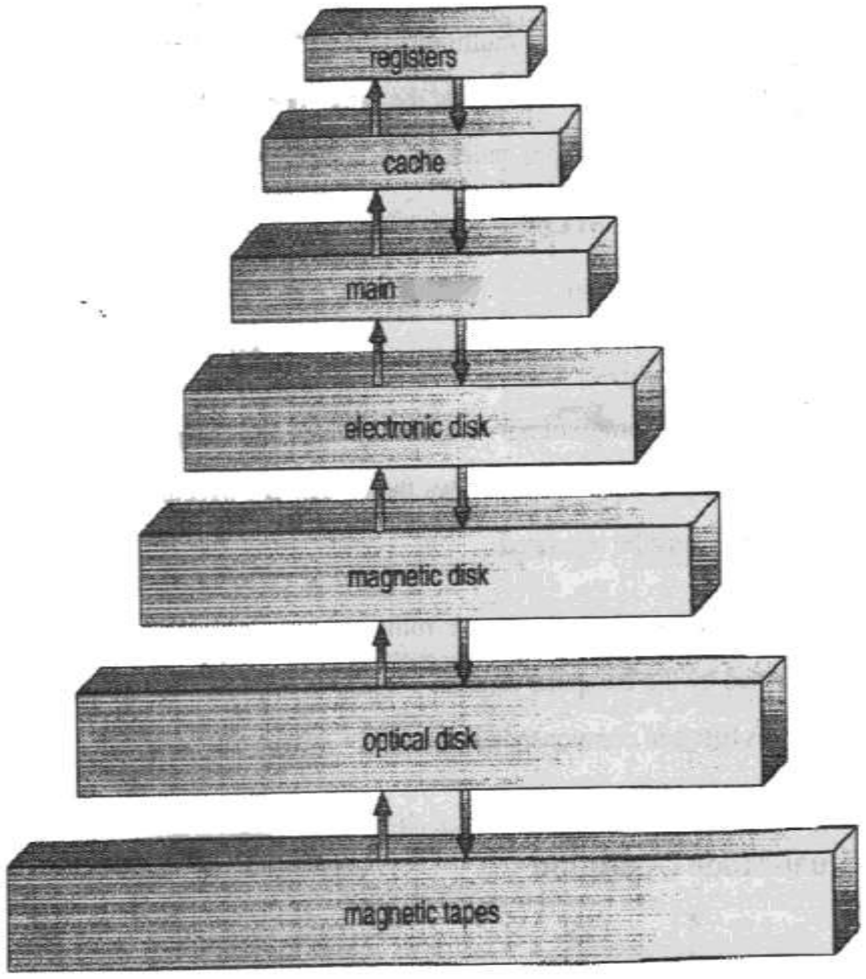
Storage Hierarchy

- The wide variety of storage systems are organised in a hierarchy according to speeds and cost. The higher levels are expensive and fast.
- In addition to having different speeds and costs, the various storage systems are either volatile or non-volatile. Volatile storage loses its contents when the power to the device is removed.
- The hierarchy is shown in the figure. In the hierarchy, the storage systems above the electronic disk are volatile, whereas those below are non-volatile.
- An electronic disk can be designed to be either volatile or non-volatile.
- Caches can be used to improve performance where a large access time or transfer rate is needed.
- Caching is an important principle of computer systems. Information is normally kept in the main memory as it is used; it is copied to the cache on a temporary basis.

STORAGE HIERARCHY

- When we need a particular piece of information, the cache is checked initially. In addition to cache, high-speed internal registers provide caching to main memory.
- Memory Storage- Device Hierarchy - Cache management is important because of the limited size of the cache. Careful selection of cache and replacement policies improves performance.
- Main memory can be viewed as a fast cache for secondary storage because the data in the secondary storage must be copied to the main memory.
- The movement of information between levels of the storage hierarchy may be either explicit or implicit. Normally, the data transfer from cache to CPU and registers is a hardware function as data transfer from disk to memory is controlled by the OS.
- In a hierarchical storage structure, the same data may appear in different levels. It will cause problems in multi-tasking, multiprocessor and distributed environments.



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CPU Protection –

- The operating system must maintain control in addition to protecting I/O and memory.
- We can use **timers** to prevent a user program from getting stuck in an infinite loop or not calling system services, and never returning control to the operating systems.
- A timer is set, to interrupt the computer after a specified period. The period may be fixed or variable.

OPERATING SYSTEM STRUCTURE

Process Management:

1. A process can be thought of as a program in execution. A program does nothing unless its instructions are executed by a CPU. A process may be a compiler, a word processor and printing output to a console.
2. A process needs certain resources like CPU time, memory files and I/O devices to accomplish the tasks. The resources are allocated when the process is created or while it is running. When the process is terminated, the operating system will reclaim reusable resources that are no longer used by that process.

Main-Memory Management

1. The main memory is central to the operation of a system. Main memory is a large array of words or bytes, ranging in size from hundreds to billions. Main memory is a repository of quick data accessible by the CPU and I/O devices.
2. The CPU reads instructions from main memory during instruction fetch and performs both read and write operations during the data fetch cycle. The I/O operations are achieved by DMA.

File Management

1. Computers can store information on several different types of physical" media. Each of these media has its own characteristics and physical organisation.
2. These properties include access speed, capacity, and data transfer rate and access method (sequential or random). So, file management is one of the components of the operating system.

OPERATING SYSTEM STRUCTURE

I/O System Management

- The operating system hides the peculiarities of specific hardware devices from the user. The I/O, subsystem consists of -
  a. A memory management component that includes buffering, caching, and spooling.
  b. A general device driver interface.
  c. Drives for specific hardware devices.

Secondary Storage Management

- The main purpose of the computer is to execute programs; hence, the programs must be in the primary memory for execution.
- The size of primary memory is small to provide all data and programs, so the system provides secondary storage to back up main memory and to store data permanently. The programs, including compilers, assemblers, sort routines, editors and formatters are stored on disk. Hence, the proper management of disk storage is important.

Command- Interpreter System

- Command interpreter is the interface between the user and the operating system.
- The MS-DOS and UNIX treat command interpreter as a special program that runs when a job is initiated. The command interpreter is often called as a shell.

OPERATING SYSTEM SERVICES

The operating system provides services to programs and to the users of those programmers to make the programming task easier. These services vary from one operating system to another.

a. **Program execution:** The program is loaded into memory for execution, and the program must end either normally or abnormally i.e. by indicating errors.
b. **I/O operations:** The operating system must provide a means to do I/O operations when a program is running. The users usually cannot control I/O devices directly for efficiency and protection.
c. **File system manipulation:** The file system reads, writes, creates and deletes files based on the program's need.
d. **Communication:** The processes need to exchange information between processes. This is done in two ways.
   The first takes place between processes that are executing on the same computer. The second takes place between processes that are executing on different computers tied by the network.
e. **Error detection:** The operating system must be aware of possible errors. Errors may occur in the CPU and Memory hardware and in I/O devices, and in the user program for each type of error, the operating system should take appropriate action for consistent computations.
f. **Resource allocation:** Resources must be allocated to processes when multiple users are logged on the system, or multiple jobs are running at the same time. The operating system provides routines to allocate plotters, modems, and other peripheral devices.

OPERATING SYSTEM SERVICES

- **Protection and Security:**
  Protection ensures that all access to system resources is controlled when several disjointed processes execute concurrently. Security of the system is also important, and usually, it is provided by means of a password to authenticate the system
- **System Calls:**
  The system call provides the interface between a process and the operating system. These calls are generally available as assembly-language instructions. Some systems allow system calls directly from a high-level language program like C, C++ and Perl.
- **Process Control:**
  A running program must need to be terminated execution either normally (end) or abnormally (abort). If a system call is made to terminate the currently running program abnormally, that causes an error trap and generates an error message along with a dump of memory. The dump is written to disk and examined by a debugger to determine the cause of the problem. The operating system must transfer control to the command interpreter under normal or abnormal circumstances.

OPERATING SYSTEM SERVICES

- **File Management**
  The file management supports various file-related system calls to create and delete files. Some system calls need file names and their attributes. Upon creation of a file, we can open it and use it to perform read, write or reposition. Finally, we should close the file.
- **Information Maintenance:**
  There are various system calls to transfer information between the user program and the operating system. Some may return information regarding user information like the current no of users, the version of the operating system, the amount of memory and so on. Along with it the operating system provides system calls to access this information.
  The various system calls supported by Information maintenance are:
  a. Get time or date, set time or date.
  b. Get system data set system data.
  c. Get process, file, or device attributes.
  d. Set process, file, or device attributes.

OPERATING SYSTEM STRUCTURE & DESIGN

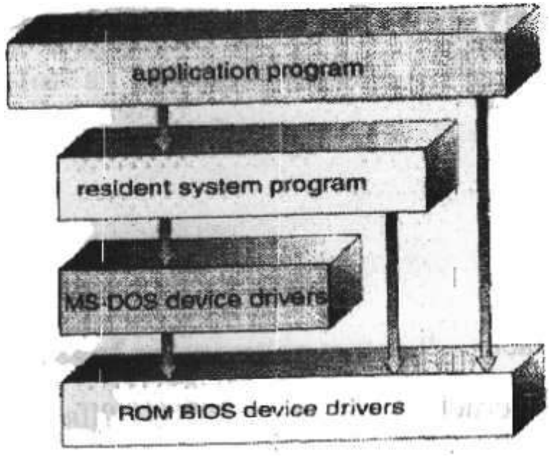
Simple structure

MS-DOS Layer

- It was written to provide the most functionality in less space, so it was not divided into layers.

In the MS-DOS. The interface and the functionality levels are not separated, so application programs can be written directly to the display and disk drives.

- Another example of limited structure is the original UNIX OS. UNIX system was initially limited by hardware functionality. It has two parts: the kernel and system programs. Everything below the system call the interface, and above the **physical** hardware is the kernel.
- The kernel provides the file system, CPU scheduling, memory management, and other operating system functions through system calls.



Microkernel

- The microkernel is an approach that modularises the kernel using an operating system called Mach. This implements components by removing all nonessential functions from the kernel and implementing them as user or system-level programs. The main function of the microkernel is to provide a communication facility between the client program and the various services that are also running in user space.
- The microkernel approach provides ease of extending the operating system. This results to port operating system, i.e. to change from one type of hardware to another. The microkernel provides more security and reliability.

SYSTEM DESIGN AND IMPLEMENTATION

- The design of a system must define the goals and specifications of the system. The design of the system will be affected by the hardware and type of the system like batch, time shared, single user, multi-user, distributed, real-time or general purpose at the high level.
- The requirements of the system are harder to specify, so requirements can be divided into two basic groups:
  1. User goals. 2. System goals

The user desires to have certain properties of the system normally as easy of use, easy to learn, reliable, safe and, fast. The similar sets of requirements are defined by the designer to design, create, maintain, and operate the system.

- **System Generation**

a. Operating systems are commonly designed to run on the class of machines at a variety of sites and a variety of peripheral configurations. So, the system must be configured or generated for each specific computer; this process is sometimes known as system generation (SYSGEN).

b. A special program is used to generate a system. The information concerning the configuration of the system is read by the SYSGEN program from a given file, or the operator of the system will provide the information.

The following kind of information is determined. Like

What CPU will be used? How much memory is available? What devices are available? What operating system options are desired, or what parameter values are to be used?

THANK YOU